

Goal: $Z_1 \dots Z_n \stackrel{iid}{\sim} P_{data}(\cdot)$

Learn $u_t^\theta(\cdot)$ s.t. $\frac{dX_t}{dt} = u_t^\theta(X_t) \quad 0 \leq t \leq 1$

$X_0 \sim P_{init}$

$\underbrace{\hspace{10em}}_A \Rightarrow X_1 \sim P_{data}$

marginal

$$\min_{\theta} \mathbb{E} \left\{ \left\| u_t^\theta(x) - \underbrace{u_t^{target}(x)}_{?} \right\|^2 \right\}$$

$t \sim U(0,1)$

$Z \sim P_{data}$

$$x \sim P_t(z) \left\{ \begin{array}{l} x \sim P_t(x|z) \end{array} \right.$$

$$u_t^{target}(x) = \int \underbrace{u_t^{target}(x|z)}_{\text{Known}} \frac{\overbrace{P_t(x|z)}^{\downarrow} \overbrace{P_{data}(z)}^{\downarrow}}{\underbrace{P_t(x)}_{\downarrow}} dz$$

$$\int P_t(x|z) P_{data}(z) dz$$

Alternative loss

$$\min_{\theta} \mathbb{E} \left\{ \left\| u_t^\theta(x) - u_t^{target}(x|z) \right\|^2 \right\}$$

$t \sim U(0,1)$
 $Z \sim P_{data}$
 $x \sim P_t(x|z)$

$$A = \mathbb{E} \left\{ \left\| u_t^\theta(x) \right\|^2 - 2 u_t^\theta(x)^T u_t^{target}(x) + \left\| u_t^{target}(x) \right\|^2 \right\}$$

$$= \mathbb{E} \left\{ \left\| u_t^\theta(x) \right\|^2 \right\} - 2 \mathbb{E} \left\{ u_t^\theta(x)^T \underbrace{u_t^{target}(x)}_{\text{dashed box}} \right\} + C$$

$$\int \int u_t^\theta(x)^T \left(\int u_t^{target}(x|z) \frac{P_t(x|z) P_{data}(z)}{P_t(x)} dz \right) \cdot \cancel{P_t(x)} dx dt$$

$$= \int \int \int u_t^\theta(x)^T u_t^{\text{target}}(x|z) P_t(x|z) \cdot P_{\text{data}}(z) dz dx dt$$

$$= \mathbb{E}_{\substack{t \sim U(0,1) \\ z \sim P_{\text{data}} \\ x \sim P_t(x|z)}} \left\{ u_t^\theta(x)^T u_t^{\text{target}}(x|z) \right\}$$

$$\begin{aligned} \Rightarrow A &= \mathbb{E} \left\{ \overbrace{\|u_t^\theta(x)\|^2}^B \right\} - 2 \mathbb{E} \left\{ \overbrace{u_t^\theta(x)^T u_t^{\text{target}}(x|z)}^D \right\} + C \\ &= \underbrace{\mathbb{E} \{ B \}}_{\substack{E \\ C' \\ C''}} - 2 \mathbb{E}(D) + \mathbb{E} \left\{ \overbrace{\|u_t^{\text{target}}(x|z)\|^2}^E \right\} - \underbrace{\mathbb{E} \{ E \}}_{C''} + C \\ &= \mathbb{E} \left\{ \|u_t^\theta(x) - u_t^{\text{target}}(x|z)\|^2 \right\} \end{aligned}$$

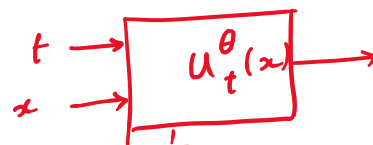
Example :

$$\begin{cases} \alpha_t = t \\ \beta_t = 1-t \end{cases}$$

$$X_t|z = \alpha_t z + \beta_t X$$

$$= t z + (1-t) X \quad X \sim \mathcal{N}(0,1)$$

$$Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} P_{\text{data}}(z)$$



$$\min_{\theta} \mathbb{E}_{\substack{t \sim U(0,1) \\ z \sim P_{\text{data}} \\ x \sim P_t(x|z)}} \left\{ \left\| u_t^\theta(x) - \underbrace{\left[\left(1 + \frac{t}{1-t}\right) z + \frac{-1}{1-t} x \right]}_{\frac{1}{1-t} [z - x]} \right\|^2 \right\}$$

$$\left(\alpha_t - \frac{\beta_t \alpha_t}{\beta_t} \right) z + \frac{\beta_t}{\beta_t} \underline{\underline{x}} = \underline{\underline{x}}$$

$$\epsilon \sim \mathcal{N}(0, 1) \leftarrow$$

$$t \sim U(0, 1)$$

$$Z \sim P_{\text{data}}(\cdot)$$

$$x = tZ + (1-t)\epsilon \rightarrow Z = \frac{t-1}{t}\epsilon + \frac{1}{t}x$$

$$Z - \epsilon = \frac{-1}{t}\epsilon + \frac{1}{t}x = \frac{1}{t}(x - \epsilon)$$

$$\text{loss} += \left\| u_t^\theta(x) - \frac{1}{1-t}(Z - \epsilon) \right\|^2$$